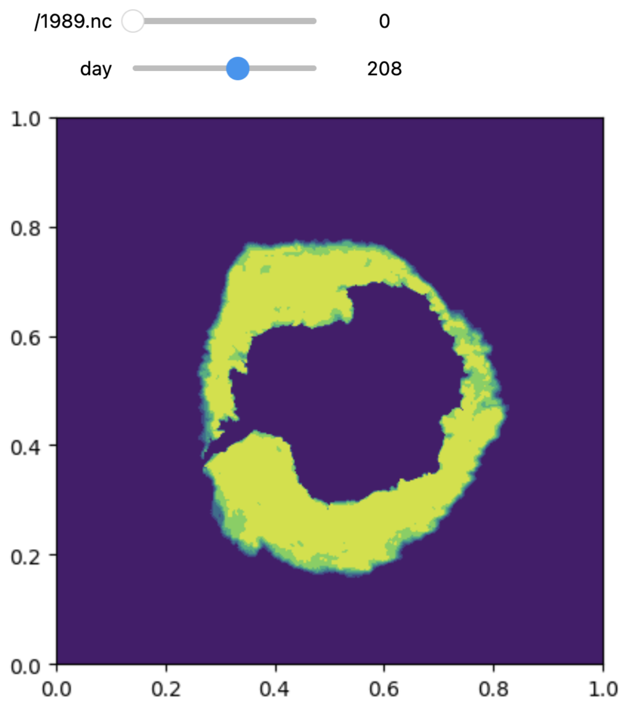
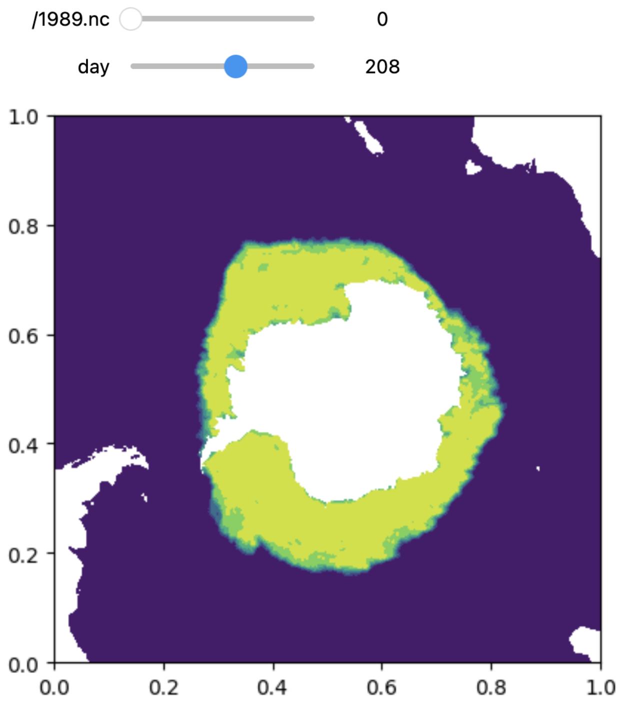
We are after predictively modelling sea-ice concentration around Antarctica, which is a scalar field with values in $[0, 1]$.

After some processing, the data can be represented as $n^t(x, y) \in [0, 1]$, where $x \in [0, 1]$, $y \in [0, 1]$, and $t \in \mathrm{N}$ represents the time, which is a discrete index.

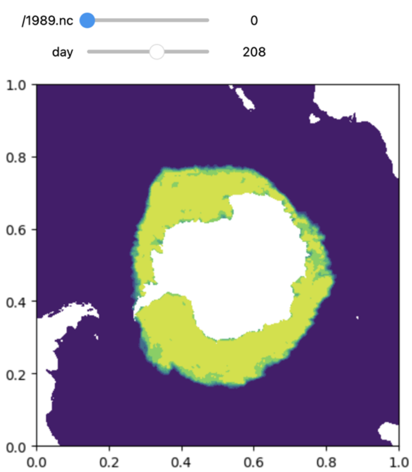
We have daily snapshots of $n^t(x, y)$ on a mesh of 432 points along x and y axes, for every day from January 1, 1989, until December 31, 2022. For example, here is day 208 of year 1989:

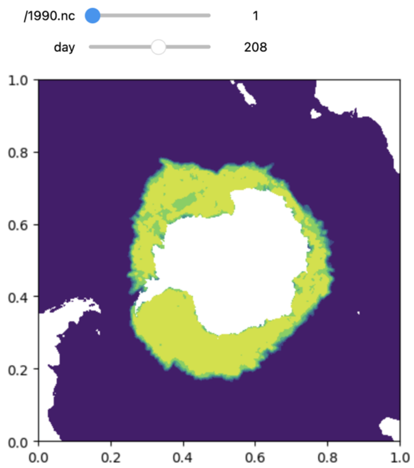


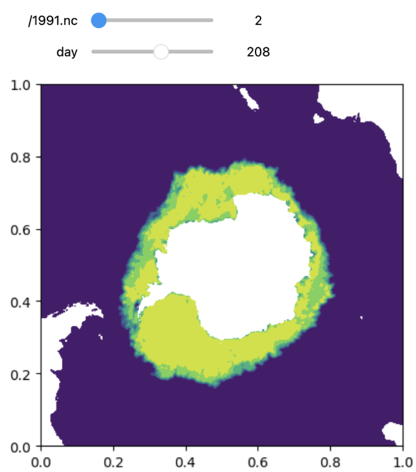
The observed value of $n(x, y)$ will always be zero on land, so we can work out where the continents intersect with the mesh and set the values of $n(x, y) := NaN$ at those points, which is only a neat plotting device. The data contains numerical values of $n(x, y)$ for any mesh point, and on land it is simply zero. But the pictures look nicer with the continental mask:

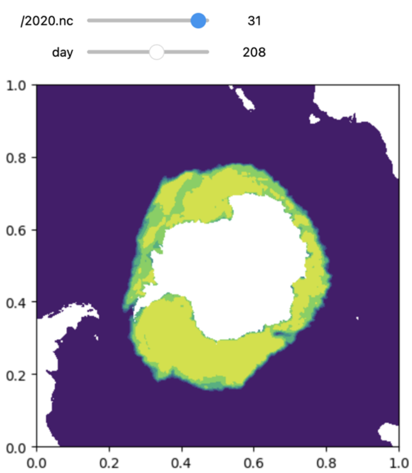


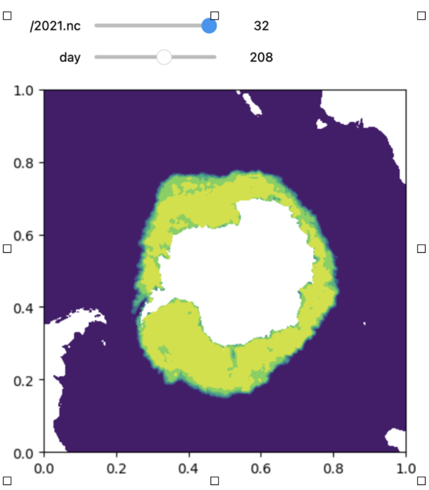
The data is periodic only "approximately", as can be seen from the following:

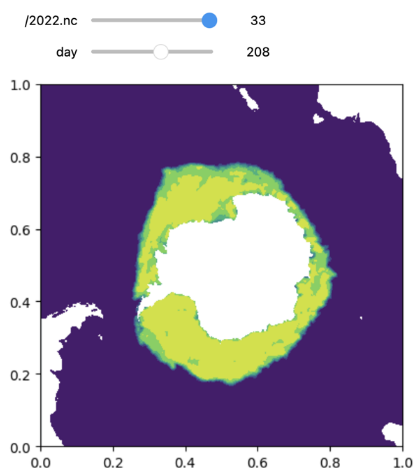










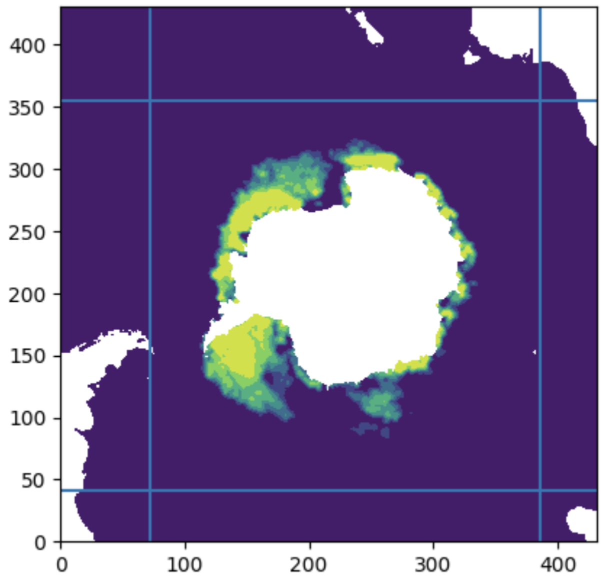


Moreover, between 1989 and 2022, climate change may have been altering the law of sea-ice dynamics.

There are 3 datasets:

Note: The field $u(x, y)$ in `full_data.pkl` and `data314.pkl` has periodic boundary conditions: it is zero on the boundaries. But not in `data64_land_1.pkl`.

1. `full_data.pkl` ~ 19Gb
   The full dataset from above. It contains variables `(continent_mask, Y, x, y)`:
   1. `continent_mask` is a Boolean array with Truth on land and False on sea
   2. `Y` is a list of 35 numpy arrays (for 35 years), where each `Y[i]` has shape `(365, 432, 432)` or `(364, 432, 432)` for day of the year and space location.
   3. `x` and `y` are equispaced mesh nodes. They are saved in the file, but are identical to `linspace(0,1,432)`
2. `data314.pkl` ~ 10 Gb
   Notice that we can truncate some of the data where $n(x, y) = 0$ by "zooming" on Antarctica. In figure below, outside of the blue square, $n(x, y) = 0$ at all days:



So we can zoom into the blue square, and rescale `x` and `y` to [0,1] each again. This results in a smaller dataset, where the mesh is `314x314` nodes, instead of 432. Also, the years are joined, and we have on continuous index `t`, indexing days.

3. `data64_land_1.pkl` ~ 0.4Gb
   Finally, for experimentation we still want a much smaller dataset. So we can zoom in to a `64x64` "representative" square and rescale x and y to [0,1]x[0,1] again: